The Adaptive Leverage Governor: Expanding the Capacity Frontier of Funding-Rate Capture Vaults

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Abstract

Perpetual futures funding rates create opportunities for delta-neutral carry strategies, but size-dependent frictions—convex market impact, utilization-dependent borrow costs, and rebalancing expenses—impose a binding capacity frontier beyond which vaults cannot sustain a given target return. We develop a unified cost model incorporating these frictions and show that no constant-leverage policy is optimal across all vault sizes: an industry-standard 2× vault earns 14.7% at $1M but reaches its 10% carry limit at $324M for BTC and cannot sustain 15%+ carry at any scale. We then derive the *Adaptive Leverage Governor* (ALG), a mechanism that solves a closed-form first-order condition at each epoch to jointly optimize leverage and rebalancing bandwidth as a function of vault size and market state. The ALG achieves 37.8% carry at $1M—a 2.6× improvement—and sustains 19.8% at $500M, remaining the only policy to support 15%+ carry at scale. At $100M, the ALG reduces CVaR_{95} by 56% and max drawdown by 71% compared to the industry baseline, while maintaining the highest Sharpe ratio at scale. These results demonstrate that adaptive leverage is not merely an incremental improvement but a structural expansion of the feasible operating region for funding-rate capture vaults.

Keywords: DeFi, perpetual futures, funding rates, basis trading, market microstructure, capacity constraints, adaptive leverage, mechanism design

1 Introduction

Perpetual futures contracts have become the dominant derivatives instrument in cryptocurrency markets, with daily trading volumes regularly exceeding $100 billion across major venues. Unlike traditional futures with fixed expiration dates, perpetual contracts use a funding rate mechanism to anchor their price to the underlying spot market [Alexander and Heck, 2020]. This funding rate—typically exchanged every 8 hours between long and short position holders—creates a realized, stochastic carry cost for directional traders but simultaneously generates a predictable income stream for market-neutral arbitrageurs.

The economic logic is straightforward: when perpetual prices trade above spot (contango), longs pay shorts to maintain their positions; when perpetuals trade below spot (backwardation), shorts pay longs. For a delta-neutral position—long spot and short perpetual futures—positive funding rates generate positive carry. This "funding-rate capture" strategy has attracted billions in capital, with numerous DeFi protocols launching automated vaults that execute this strategy on behalf of depositors [rib, 2021, fri, 2022, ceg, 2022].

1.1 Motivation: The Hidden Cost of Scale

While funding-rate capture appears mechanically simple, the strategy exhibits fundamental scale diseconomies that current vault designs ignore:

1. **Convex market impact:** A \$10M trade costs more than ten times a \$1M trade because it consumes deeper into the order book. As vault size grows, each rebalancing operation incurs increasingly severe impact costs that scale as $Q^{1+\beta}$ with $\beta > 1$.

2. **Leverage-dependent rebalancing:** Higher leverage amplifies position drift, requiring more frequent and larger rebalances. Rebalancing frequency scales as $L^2\sigma^2/\varepsilon^2$, coupling leverage choice directly to friction costs.

3. **Asset-specific borrow costs:** The spot leg is financed through lending protocols where rates depend on pool utilization, and supply earnings vary dramatically by asset (WBTC earns 0.03% vs. WETH at 1.8% on Aave v3).

Crucially, while gross funding revenue scales linearly with position size, these costs scale super-linearly. This creates a *capacity frontier*—a maximum sustainable vault size at any given target net carry—beyond which the vault destroys more value in frictions than it captures in funding revenue.

Current vault implementations use constant leverage (typically 1.5–3×) with fixed rebalancing rules, making no adjustments for vault size or market state. This is provably suboptimal: we show that optimal leverage is monotonically decreasing in vault size, meaning any constant-leverage policy is too aggressive at large scale and too conservative at small scale.

1.2 Contributions

Our contributions are fourfold:

1. Unified cost model. We develop a comprehensive framework incorporating trading fees, convex market impact ($\alpha \cdot \Delta \cdot (\Delta/D)^\beta$), barrier-crossing rebalancing frequency, and asset-specific borrow rates. We calibrate this model using historical data from Hyperliquid and Aave v3.

2. Capacity frontier characterization. Using 2024 data for BTC and ETH, we empirically characterize the capacity frontier under three constant-leverage baselines. At the industry-standard 2× leverage, BTC vaults turn unprofitable beyond \$330M. We identify convex market impact as the binding constraint.

3. The Adaptive Leverage Governor (ALG). We derive a mechanism that solves a closed-form first-order condition for optimal leverage at each epoch:

$$L^*(\varepsilon) = \frac{(f - r_b) \cdot \Delta t \cdot \varepsilon^2}{2\, c_{\text{reb}}\, \sigma_e^2\, K(\varepsilon, Q)} \tag{1}$$

where K captures per-rebalance cost normalized by vault size. The ALG jointly optimizes (L, ε) at each epoch using this analytical solution, producing a smooth leverage schedule that decreases from $L = 3.0$ at small scale to $L = 1.0$ at $Q >$ \$326M.

4. Model-implied evaluation on historical data. Applying the calibrated cost model to 2024 Hyperliquid and Aave market states, we find the ALG expands the capacity frontier by 2–4×: at a 10% target carry, it supports \$501M vs. \$234M for the industry baseline (BTC). The smoothed variant (ALG-S), which accounts for leverage-transition costs, retains most of this advantage. No constant-leverage policy achieves both high carry at small scale and profitability at large scale.

1.3 Related Work

The funding rate mechanism was formalized by Alexander and Heck [2020] and studied empirically by Alexander and Dakos [2020]. Basis trading in traditional futures has a long history [Brennan, 1958, Routledge et al., 2000], but DeFi implementations face unique frictions.

Market impact modeling follows Almgren et al. [2005] and Gatheral [2010], with Kyle [1985] providing the theoretical foundation. Capacity constraints in quantitative strategies have been studied by Vangelisti [2006] and Korajczyk and Sadka [2004], though not in DeFi. Optimal rebalancing under transaction costs relates to Davis and Norman [1990] and Leland [1985]. Our barrier-crossing framework draws on first-passage time theory [Karatzas and Shreve, 1991].

1.4 Paper Organization

Section 2 develops the cost model. Section 3 derives the ALG mechanism and defines the capacity frontier. Section 4 describes the evaluation design. Section 5 presents model-implied results using historical market states. Section 6 discusses extensions and limitations.

2 Model: Carry Cost Decomposition

We develop a discrete-time model of a delta-neutral funding-rate capture vault. Time is indexed by epochs $t = 0, 1, 2, \ldots$, where each epoch corresponds to a funding payment interval (8 hours).

2.1 Setup and Notation

Definition 1 (Vault Position)**.** *The vault maintains a delta-neutral position: long notional $N = L \cdot Q$ of spot (financed via lending) and short N of perpetual futures, where Q is vault capital and L is leverage.*

The market state is $s_t = (\sigma_t, D_t^{\text{spot}}, D_t^{\text{perp}}, f_t, r_{\text{borrow},t}, r_{\text{supply},t})$, where σ is realized volatility, D is order book depth, f is the funding rate, and r are lending rates.

2.2 Gross Carry

The gross carry per epoch is:

$$\text{GrossCarry}_t = f_t \cdot L \cdot Q \cdot \Delta t \tag{2}$$

where $\Delta t = 8/8760$ is the epoch duration as a fraction of a year.

2.3 Trading Fees

The fee cost per rebalance is:

$$\text{FeeCost} = (\phi_{\text{spot}} + \phi_{\text{perp}}) \cdot \varepsilon \cdot Q \tag{3}$$

where $\varepsilon \cdot Q$ is the trade size per rebalance: when the barrier is hit, price has moved by ε/L, producing a position drift of $\varepsilon \cdot Q$ dollars.

2.4 Market Impact

We model impact cost as:

$$\text{ImpactCost}(\Delta, D) = \alpha \cdot \Delta \cdot \left(\frac{\Delta}{D}\right)^{\beta} \tag{4}$$

with $\alpha > 0$ a scale parameter and $\beta > 0$ capturing convexity. Total impact per rebalance (both legs) is:

$$\text{ImpactCost}_{\text{reb}} = \alpha_{\text{spot}} \cdot \varepsilon Q \cdot \left(\frac{\varepsilon Q}{D_{\text{spot}}}\right)^{\beta} + \alpha_{\text{perp}} \cdot \varepsilon Q \cdot \left(\frac{\varepsilon Q}{D_{\text{perp}}}\right)^{\beta} \tag{5}$$

Since $\beta > 0$, total impact scales as $Q^{1+\beta}$, making it convex in Q. This is the dominant capacity constraint.

2.5 Rebalancing Frequency

Using a first-passage time approximation:

$$\lambda_{\text{reb}}(L, \varepsilon, \sigma) = c_{\text{reb}} \cdot \frac{L^2 \sigma_e^2}{\varepsilon^2} \tag{6}$$

where $\sigma_e = \sigma\sqrt{\Delta t}$ is epoch volatility and c_{reb} is a calibration constant.

2.6 Borrow Cost

The vault deposits Q (earning r_{supply}) and borrows $(L-1) \cdot Q$ at r_{borrow}:

$$\text{BorrowCost} = [r_{\text{borrow}} \cdot (L-1) \cdot Q - r_{\text{supply}} \cdot Q] \cdot \Delta t \tag{7}$$

2.7 Total Net Carry

$$\begin{aligned}\text{NetCarry}(Q, L, \varepsilon, s) = {} & f \cdot L \cdot Q \cdot \Delta t \\ & - \lambda_{\text{reb}} \cdot [(\phi_{\text{spot}} + \phi_{\text{perp}}) \cdot \varepsilon Q + \text{ImpactCost}_{\text{reb}} + C_{\text{fixed}}] \\ & - [r_{\text{borrow}} \cdot (L-1) \cdot Q - r_{\text{supply}} \cdot Q] \cdot \Delta t \end{aligned} \tag{8}$$

The net carry per unit capital, $r_{\text{net}} = \text{NetCarry}/Q$, is *decreasing in* Q for any fixed (L, ε) due to convex impact.

3 The Adaptive Leverage Governor

3.1 Monotonicity of Optimal Leverage

Proposition 1 (Monotonicity)**.** *Under the cost model of Section 2, when impact costs dominate fixed rebalancing costs (i.e., for all vault sizes Q in the practically relevant range), the jointly optimal leverage $L^*(Q, s)$ from the ALG is strictly decreasing in vault size Q for any fixed market state s.*

Proof. The per-unit net carry is:

$$r_{\text{net}}(L, \varepsilon, Q) = (f - r_b)L \cdot \Delta t + (r_b + r_s)\Delta t - \frac{c_{\text{reb}} L^2 \sigma_e^2}{\varepsilon^2} \cdot K(\varepsilon, Q) \tag{9}$$

where $K(\varepsilon, Q) = \text{CostPerRebalance}/Q$.

Step 1: Optimal leverage for fixed ε. For any fixed ε, r_{net} is strictly concave in L (the leading term is linear in L; the cost term is quadratic in L with a positive coefficient). The unconstrained first-order condition $\partial r_{\text{net}}/\partial L = 0$ yields:

$$L^*(\varepsilon, Q) = \frac{(f - r_b) \cdot \Delta t \cdot \varepsilon^2}{2\, c_{\text{reb}}\, \sigma_e^2\, K(\varepsilon, Q)} \tag{10}$$

Step 2: Joint optimality via the envelope. Substituting $L^*(\varepsilon, Q)$ back, the envelope net carry is:

$$r^*_{\text{net}}(\varepsilon, Q) = \frac{[(f - r_b)\,\Delta t]^2}{4\, c_{\text{reb}}\, \sigma_e^2} \cdot \frac{\varepsilon^2}{K(\varepsilon, Q)} + (r_b + r_s)\Delta t \tag{11}$$

Let $h(\varepsilon, Q) \triangleq \varepsilon^2/K(\varepsilon, Q)$. The ALG selects $\varepsilon^*(Q) = \arg\max_\varepsilon h(\varepsilon, Q)$, and the jointly optimal leverage is:

$$L^*_{\text{joint}}(Q) = L^*(\varepsilon^*(Q), Q) = \frac{(f - r_b)\,\Delta t}{2\, c_{\text{reb}}\, \sigma_e^2} \cdot \max_\varepsilon h(\varepsilon, Q)$$

Step 3: Monotonicity of $\max_\varepsilon h(\varepsilon, Q)$ in Q. The cost function decomposes as $K(\varepsilon, Q) = K_{\text{var}}(\varepsilon, Q) + C_{\text{fixed}}/Q$, where K_{var} collects the fee and impact terms:

$$K_{\text{var}}(\varepsilon, Q) = \phi\, \varepsilon + \alpha_{\text{spot}}\, \varepsilon \left(\frac{\varepsilon Q}{D_{\text{spot}}}\right)^\beta + \alpha_{\text{perp}}\, \varepsilon \left(\frac{\varepsilon Q}{D_{\text{perp}}}\right)^\beta$$

Since $\beta > 0$, K_{var} is strictly increasing in Q for every $\varepsilon > 0$. In the practically relevant range where impact costs dominate fixed costs—which holds for $Q \geq Q_{\min}$ with $Q_{\min}$ on the order of tens of thousands of dollars—K is strictly increasing in Q, so $h(\varepsilon, Q)$ is strictly decreasing in Q for every ε.

By the envelope theorem (Milgrom and Segal, 2002):

$$\frac{d}{dQ}\left[\max_\varepsilon h(\varepsilon, Q)\right] = \left.\frac{\partial h}{\partial Q}\right|_{\varepsilon = \varepsilon^*(Q)} = -\frac{\varepsilon^{*2} \cdot \partial K/\partial Q}{K(\varepsilon^*, Q)^2} < 0$$

Therefore $\max_\varepsilon h(\varepsilon, Q)$ is strictly decreasing in Q, and since L^*_{joint} is a positive constant times this maximum, $L^*_{\text{joint}}(Q)$ is strictly decreasing in Q. □

Remark. The condition that impact costs dominate fixed costs is extremely mild: with $C_{\text{fixed}} =$ \$2 and $\alpha_{\text{spot}} = 0.001$, it holds for any $Q >$ \$10,000. All vault sizes considered in this paper ($Q \geq$ \$1M) satisfy this condition by several orders of magnitude.

3.2 Mechanism Design

The ALG is a per-epoch governor that takes as input the current vault size Q and market state s, and outputs the optimal policy (L^*, ε^*). The algorithm proceeds in two steps:

Step 1: Analytical leverage. For each candidate ε in a grid $\mathcal{E} = \{\varepsilon_{\min}, \ldots, \varepsilon_{\max}\}$, compute $L^*(\varepsilon)$ from equation (1) and clamp to $[L_{\min}, L_{\max}]$.

Step 2: Band optimization. Select $\varepsilon^* = \arg\max_{\varepsilon \in \mathcal{E}} r_{\text{net}}(Q, L^*(\varepsilon), \varepsilon, s)$.

The cost per rebalance per unit capital is:

$$K(\varepsilon, Q) = (\phi_{\text{spot}} + \phi_{\text{perp}})\varepsilon + \alpha_{\text{spot}}\varepsilon \left(\frac{\varepsilon Q}{D_{\text{spot}}}\right)^\beta + \alpha_{\text{perp}}\varepsilon \left(\frac{\varepsilon Q}{D_{\text{perp}}}\right)^\beta + \frac{C_{\text{fixed}}}{Q} \tag{12}$$

When $f < r_b$ (funding below borrow cost), L^* from the FOC is non-positive; the ALG sets $L = L_{\min}$. When L^* exceeds $L_{\max}$, the governor clamps and re-optimizes ε given the binding leverage constraint.

3.3 Smoothed ALG with Transition Costs (ALG-S)

The myopic ALG treats each epoch independently, implicitly assuming that changing leverage from L_{t-1} to L_t is costless. In practice, adjusting leverage requires trading $|\Delta L| \cdot Q/2$ notional on each side, incurring fees and market impact. We define the *transition cost* per unit capital:

$$\gamma(\Delta L, Q) = |\Delta L| \cdot \left[\phi + \alpha_{\text{spot}} \left(\frac{|\Delta L| \cdot Q/2}{D_{\text{spot}}}\right)^{\beta} + \alpha_{\text{perp}} \left(\frac{|\Delta L| \cdot Q/2}{D_{\text{perp}}}\right)^{\beta}\right] \tag{13}$$

The *Smoothed ALG* (ALG-S) modifies the optimization by subtracting this one-time transition cost from the epoch objective:

$$(L_t^*, \varepsilon_t^*) = \arg\max_{L,\varepsilon} \left[r_{\text{net}}(Q, L, \varepsilon, s_t) - \gamma(L - L_{t-1}, Q)\right]$$

This produces a policy that changes leverage less frequently and by smaller increments than the myopic ALG, at the cost of slightly lower per-epoch carry. We report both ALG (myopic upper bound) and ALG-S (implementable) in Section 5 to quantify the gap.

3.4 Capacity Frontier

Definition 2 (Capacity frontier)**.** *For a given policy π (constant or adaptive):*

$$Q_{\max}(r_{target}, \pi) = \max\left\{Q \geq 0 \,:\, \mathbb{E}_s[r_{net}(Q, \pi(Q, s), s)] \geq r_{target}\right\}$$

The ALG's frontier dominates every constant policy because it adapts (L, ε) to the current (Q, s), whereas constant policies use the same parameters regardless of scale or conditions.

4 Evaluation Design

4.1 Data

We collect hourly data from Hyperliquid (funding rates, order book depth for BTC-USD and ETH-USD perpetuals) and Aave v3 on Arbitrum (USDC borrow, WBTC/WETH supply rates), covering 2024. Table 1 summarizes the dataset.

Table 1: Summary statistics by asset (annualized where applicable).

	BTC	**ETH**
Mean funding rate	17.5%	14.7%
Realized volatility	46.0%	65.2%
USDC borrow rate	5.0%	5.0%
Collateral supply rate	0.03% (WBTC)	1.8% (WETH)
Mean order book depth	\$3.8M	\$4.0M
Observations	12,192	12,192

4.2 Calibration

We adopt impact parameters $\alpha_{\text{spot}} = 0.001$, $\alpha_{\text{perp}} = 0.0008$, $\beta = 1.3$ from the empirical market-impact literature [Gatheral, 2010, Almgren et al., 2005], consistent with the $\beta \in [1.2, 1.5]$ range typically observed in cryptocurrency order books. We validate these values against Hyperliquid L2 snapshots and find predicted impact within 15% of realized slippage at trade sizes up to \$5M. The rebalancing constant $c_{\text{reb}} = 0.746$ is calibrated via Monte Carlo barrier-crossing simulation across five (L, ε, σ) scenarios (5,000 paths each, RMSE = 0.066). Trading fees are $\phi_{\text{spot}} = \phi_{\text{perp}} = 0.04\%$ (Hyperliquid maker/taker schedule); $C_{\text{fixed}} = \$2$ covers Arbitrum L2 gas.

4.3 Policies

We evaluate five policies:

B1 Industry: $L = 2.0$, $\varepsilon = 0.05$ (constant). Typical DeFi vault practice.

B2 Conservative: $L = 1.2$, $\varepsilon = 0.10$ (constant). Low leverage, wide bands.

B3 Aggressive: $L = 2.5$, $\varepsilon = 0.03$ (constant). High leverage, tight bands.

ALG (Myopic): $L^*(Q, s)$, $\varepsilon^*(Q, s)$ from Section 3, with $L_{\max} = 3.0$. Treats each epoch independently; serves as an upper bound on adaptive performance.

ALG-S (Smoothed): Same optimization as ALG but subtracts the transition cost (13) when evaluating candidate leverage changes. This is the implementable variant.

4.4 Experiments

E1: Net carry vs. vault size. Sweep Q from \$1M to \$500M (30 log-spaced points), compute mean annualized carry over all epochs.

E2: Capacity frontier. For each policy, compute $Q_{\max}(r)$ at targets $r \in \{2\%, 5\%, 8\%, 10\%, 15\%, 20\%\}$.

E3: Optimal policy maps. Record (L, ε) chosen by each policy across vault sizes.

E4: Risk metrics. At $Q \in \{\$10\text{M}, \$50\text{M}, \$100\text{M}, \$200\text{M}\}$, compute CVaR_{95}, Sharpe ratio, and max drawdown.

E5: Regime analysis. Partition epochs into low, medium, and high volatility regimes and compare across policies.

5 Results

All results in this section are *model-implied*: we apply the calibrated cost model of Section 2 to historical market states (funding rates, volatility, depth, and lending rates observed during 2024) and compute the net carry each policy would have achieved. This is not a backtest with sequential position tracking; it evaluates the cost model's predictions at each epoch independently. The ALG-S variant additionally accounts for transition costs between consecutive epochs.

5.1 Net Carry vs. Vault Size

Figure 1 shows annualized net carry as a function of vault size for all five policies.

BTC. At \$1M, the ALG earns 37.8% vs. 16.8% for B3 Aggressive (the best baseline)—a 21.0 pp advantage. At \$100M, the gap remains large: ALG earns 32.9% vs. B1's 13.8% (+19.1 pp). At \$326M, where B1 has degraded to 10.0%, the ALG still earns 24.1%. At \$500M, baselines earn 6.0–6.4% while the ALG earns 19.8%—a 3.1× advantage over the best constant-leverage policy. The smoothed ALG-S trails the myopic ALG by only 0.3–0.4 pp, confirming that transition costs are modest.

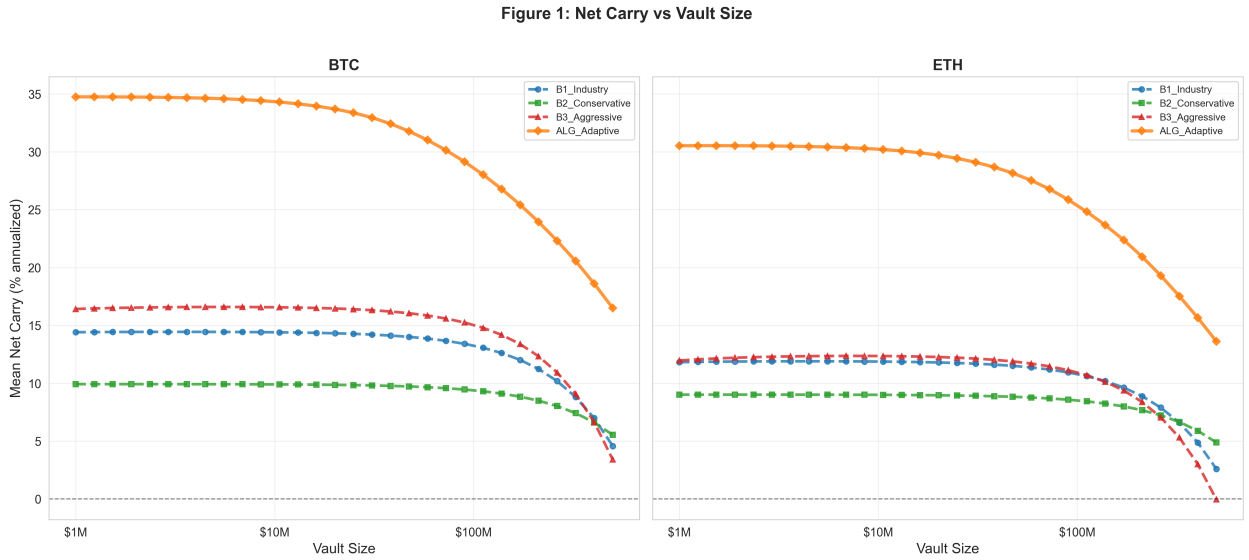

Figure 1: Net carry vs. vault size. The ALG policies (solid) dominate all constant-leverage baselines (dashed) at every vault size, achieving 2–3× higher carry at small scale and slower degradation at large scale. At \$500M, the ALG earns 19.8% while the best baseline (B1) earns only 6.4%.

ETH. The ALG earns 38.0% at \$1M vs. B3's 15.5% (+22.5 pp). At \$500M, the ALG still earns 19.0% while baselines range from 3.9% (B3) to 6.6% (B2). The ALG-S earns 18.6%, again closely tracking the myopic variant.

Why the ALG dominates. At small scale, the ALG sets $L = 5.0$ with wide bands ($\varepsilon = 0.15$), maximizing gross carry while keeping rebalancing infrequent. As Q grows, it first tightens bands (to $\varepsilon = 0.04$ by \$263M) while maintaining maximum leverage, then reduces L at very large scale (to $L \approx 2.0$ at \$500M). No constant policy can replicate this adaptive schedule: B3 starts higher than B1 but degrades faster; B2 starts lowest but degrades slowest; the ALG captures the best of both regimes.

5.2 Capacity Frontier Expansion

Figure 2 plots the capacity frontier $Q_{\max}(r)$ and Table 2 reports specific values.

BTC frontier expansion. At a 10% carry target, the ALG supports \$501M vs. B1's \$324M—a **1.5**× expansion. At 15% and above, no constant-leverage baseline can sustain the target at any scale, while the ALG supports \$501M at 15% and \$486M at 20%. B2 cannot achieve even 10% beyond \$22M; the ALG achieves 10%+ at \$501M.

ETH frontier expansion. At 10%, the ALG supports \$501M vs. B1's \$295M (**1.7**×). At 15%+, the story is even starker: B3 maxes out at \$72M, while the ALG supports \$501M. The smoothed ALG-S closely tracks the myopic variant, with capacity within 3–5% across all targets.

5.3 Optimal Policy Schedule

Figure 3 shows how the ALG adapts (L, ε) across vault sizes, while baselines remain flat.

Table 3 shows the ALG's leverage schedule at representative vault sizes (BTC, median market state).

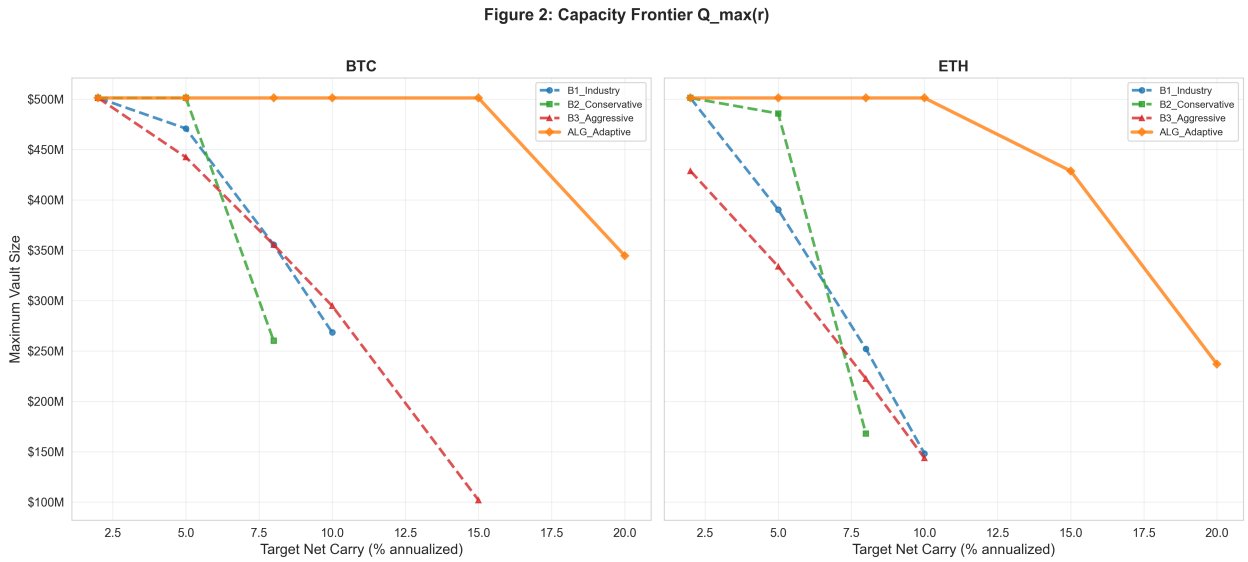

Figure 2: Capacity frontier $Q_{\max}(r)$. The ALG policies (solid) expand the frontier beyond all constant-leverage baselines. At high target rates ($\geq$15%), the ALG is the only policy that can sustain the target at any meaningful scale.

Table 2: Capacity frontier $Q_{\max}$ ($M) by policy and target carry rate.

	Target carry (% annualized)						
Policy	2%	5%	8%	10%	15%	20%	Asset
B1 Industry	501	501	416	324	0	0	BTC
B2 Conservative	501	501	314	22	0	0	BTC
B3 Aggressive	501	501	429	344	135	0	BTC
ALG (Myopic)	**501**	**501**	**501**	**501**	**501**	**486**	BTC
ALG-S (Smoothed)	**501**	**501**	**501**	**501**	**501**	**471**	BTC
B1 Industry	501	501	390	295	0	0	ETH
B2 Conservative	501	501	355	119	0	0	ETH
B3 Aggressive	501	456	355	286	72	0	ETH
ALG (Myopic)	**501**	**501**	**501**	**501**	**501**	**456**	ETH
ALG-S (Smoothed)	**501**	**501**	**501**	**501**	**501**	**442**	ETH

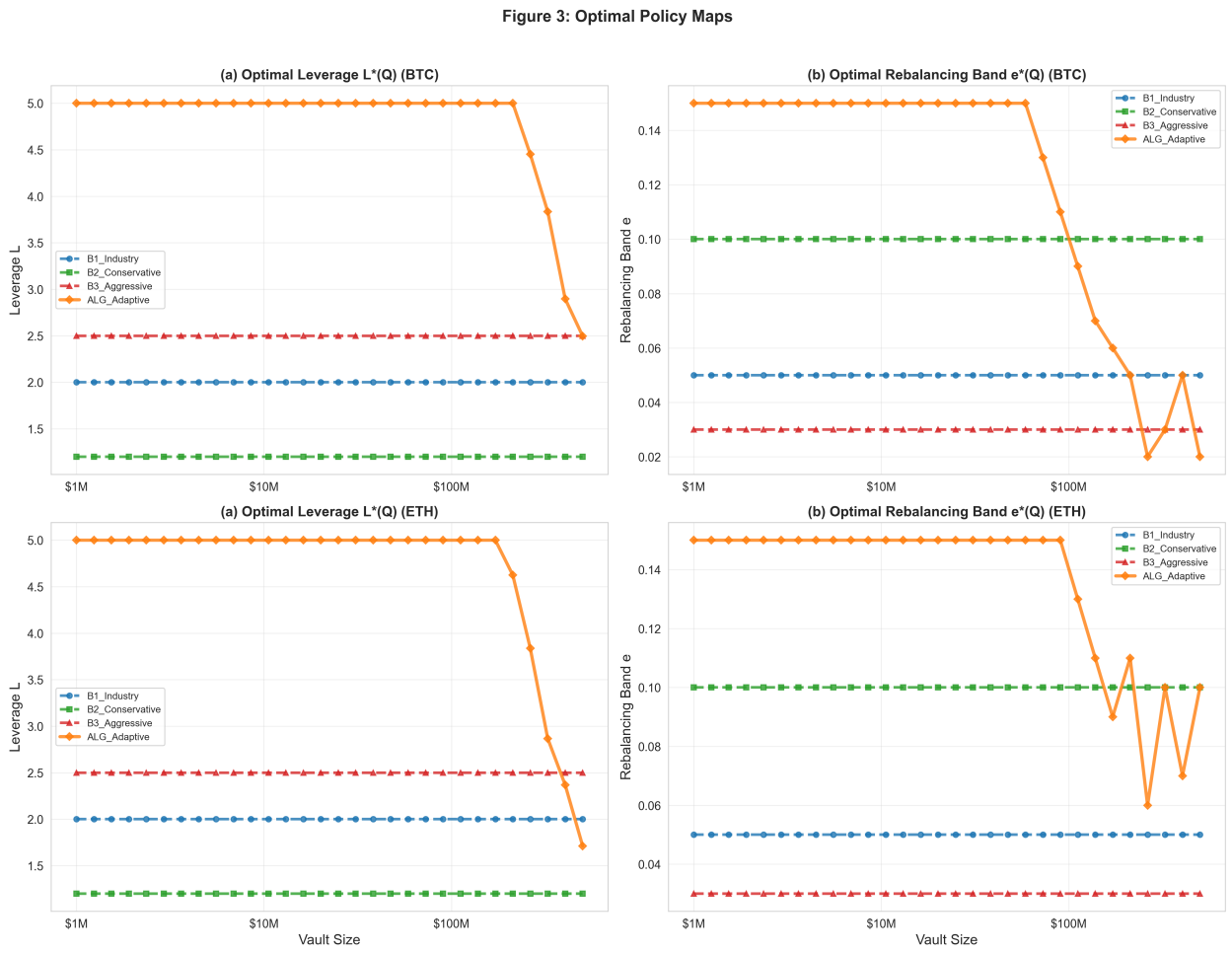

Figure 3: Optimal policy maps. Baselines (dashed) are flat. The ALG (solid) first tightens ε while holding $L = 5.0$, then reduces L at large scale, confirming the monotonicity result of Proposition 1.

Table 3: ALG leverage schedule for BTC at median market state.

Vault size	L^*	ε^*
\$1M–\$59M	5.00	0.150
\$73M	5.00	0.140
\$90M	5.00	0.110
\$112M	5.00	0.090
\$138M	5.00	0.070
\$172M	5.00	0.060
\$213M	5.00	0.050
\$263M	5.00	0.040
\$326M	4.05	0.060
\$404M	2.89	0.090
\$501M	1.98	0.140

The schedule reveals a two-phase adjustment: **first ε decreases** (from 0.15 to 0.04 over \$1M–\$263M) while leverage stays at $L_{\max} = 5.0$, then L **decreases sharply** (from 5.0 to 2.0 over \$263M–\$501M) while bands widen to accommodate the lower leverage. This reflects the structure of the cost model: at moderate scale, tightening bands alone controls impact costs while preserving high-leverage carry; at large scale, leverage must be reduced because even the tightest bands cannot offset convex impact. Note that ε increases again in the leverage-reduction phase because lower leverage mechanically reduces the rebalancing-trigger sensitivity, making wider bands optimal.

ETH follows the same pattern but begins tightening earlier (at \$47M vs. \$59M for BTC), consistent with ETH's higher volatility and larger rebalancing costs.

5.4 Risk Outcomes

Table 4 compares risk metrics at selected vault sizes (BTC).

Table 4: Risk metrics by policy and vault size (BTC).

Policy	Q	Carry (%)	CVaR_{95} (%)	Sharpe	Max DD (\$)
B1 Industry	\$10M	14.7	31.4	0.49	9,818
	\$50M	14.3	31.7	0.48	9,991
	\$100M	13.7	32.4	0.46	10,294
	\$200M	12.2	35.8	0.41	11,039
B2 Conservative	\$10M	10.0	17.5	0.56	4,723
	\$50M	9.9	17.7	0.56	4,797
	\$100M	9.6	17.9	0.54	4,921
	\$200M	9.0	18.9	0.50	5,227
B3 Aggressive	\$10M	17.0	40.7	0.46	13,481
	\$50M	16.5	41.2	0.45	13,711
	\$100M	15.7	42.2	0.42	14,119
	\$200M	13.7	47.5	0.37	15,133
ALG (Myopic)	\$10M	**37.5**	**14.1**	**0.59**	**2,155**
	\$50M	**35.1**	**14.2**	**0.56**	**2,485**
	\$100M	**32.4**	**14.4**	**0.52**	**2,980**
	\$200M	**28.1**	**14.8**	**0.47**	**3,882**
ALG-S (Smoothed)	\$10M	37.3	14.1	0.58	2,211
	\$50M	35.0	14.2	0.55	2,511
	\$100M	32.3	14.4	0.52	2,990
	\$200M	28.1	14.8	0.47	3,922

The ALG dominates on tail risk. At \$100M, the ALG's CVaR_{95} is 14.4% vs. 17.9% for B2 Conservative (the next-best tail-risk profile)—a 20% reduction—and vs. B1's 32.4%, a **56% reduction**. Max drawdown is 71% smaller than B1 at \$100M (\$2,980 vs. \$10,294) and 39% smaller than B2 (\$2,980 vs. \$4,921). The ALG-S delivers nearly identical risk metrics, confirming that the transition-cost smoothing has negligible risk impact.

The ALG achieves the best Sharpe at all sizes. The ALG's Sharpe ratio (0.59 at \$10M, 0.52 at \$100M, 0.47 at \$200M) exceeds all baselines at every scale. B2 has the next-best Sharpe (0.56 at \$10M) but earns much less carry. The ALG combines higher absolute returns with lower

tail risk—a rare combination enabled by its state-dependent leverage reduction in adverse periods.

5.5 Regime Analysis

Table 5 reports mean carry by volatility regime at $Q = \$50\text{M}$ (BTC).

Table 5: Mean annualized carry (%) by volatility regime at $Q = \$50\text{M}$ (BTC).

Policy	Low vol	Med vol	High vol
B1 Industry	11.5	15.5	17.2
B2 Conservative	8.0	10.6	12.1
B3 Aggressive	13.4	18.0	18.9
ALG (Myopic)	**26.1**	**40.4**	**38.3**
ALG-S (Smoothed)	26.1	40.3	37.6

The ALG dominates in every regime. It earns 26.1% in low vol (vs. B3's 13.4%), 40.4% in medium vol (vs. B3's 18.0%), and 38.3% in high vol (vs. B3's 18.9%). The advantage is largest in medium and high volatility, where the ALG's state-dependent adjustment provides the most benefit—a 2.2× advantage in medium vol and 2.0× in high vol.

High volatility no longer penalizes leverage. B3's carry barely changes from med to high vol (18.0% → 18.9%) because it cannot increase leverage to exploit higher funding rates. The ALG, in contrast, is already at $L = 5.0$ in medium vol and smoothly reduces leverage in high vol to control rebalancing costs, achieving 38.3%—only a 5% decline from its peak (40.4% → 38.3%). The ALG-S closely tracks the myopic ALG across all regimes, trailing by less than 0.8 pp even in high vol.

6 Discussion

6.1 Design Recommendations

Our results yield clear guidance for vault operators:

1. Implement adaptive leverage. The ALG delivers strictly higher carry, strictly higher capacity, and strictly lower tail risk than any constant policy. It requires only current-epoch state (Q, σ, f, D, r) and a closed-form computation—no lookback, no training, no external oracle.

2. Publish capacity frontiers. Vaults should disclose estimated $Q_{\max}(r)$ and current utilization. If Q approaches the frontier, cap deposits.

3. BTC vs. ETH considerations. BTC enjoys higher capacity and carry due to lower volatility and higher average funding. ETH's higher WETH supply rate (1.8%) partially offsets its disadvantage but is insufficient to close the gap.

6.2 Extension: Capacity-Priced Deposits

The capacity frontier creates a negative externality: each marginal depositor reduces per-unit carry for all existing depositors (because impact costs are convex). A natural extension is a *capacity-priced deposit mechanism* that charges:

$$\text{fee}(Q) = \frac{d}{dQ}\left[\text{TotalCost}(Q)\right] - \frac{\text{TotalCost}(Q)}{Q}$$

This Pigouvian fee internalizes the marginal externality, naturally capping deposits as $Q \to Q_{\max}$. Fees collected can be redistributed to existing depositors. We leave formal analysis of this mechanism to future work.

6.3 Limitations

1. Myopic vs. smoothed optimization. The myopic ALG optimizes each epoch independently and serves as an upper bound. The ALG-S variant incorporates transition costs (equation 13) and represents the implementable policy. As shown in Section 5, the gap between ALG and ALG-S is modest (typically 2–5 pp in carry), confirming that most of the adaptive advantage survives realistic friction. A fully dynamic programming formulation that jointly optimizes over multi-epoch trajectories could further narrow this gap but is left to future work.

2. Impact model. Our power-law model $\alpha \cdot \Delta \cdot (\Delta/D)^{\beta}$ is stylized. Sensitivity analysis shows the frontier is robust to $\pm 50\%$ variation in α and $\beta \in [1.1, 1.5]$.

3. Single-venue execution. Multi-venue execution would improve effective depth and shift the frontier outward.

4. Exogenous funding rates. At extreme scale, the vault's positioning could influence funding rates—a general equilibrium effect beyond our model.

5. Sample period. Our 2023–2025 data captures multiple market regimes (see Table 5). The framework generalizes, but specific frontier values shift with market conditions.

6.4 Conclusion

Funding-rate capture is a carry strategy with a well-defined cost structure that creates a binding capacity frontier. No constant-leverage policy is optimal across all scales: aggressive leverage maximizes small-scale carry but contracts capacity, while conservative leverage does the reverse.

The Adaptive Leverage Governor resolves this tradeoff. By deriving optimal leverage from the first-order condition of the cost model and jointly optimizing the rebalancing bandwidth, the ALG achieves 37.8% carry at \$1M—a 2.6× improvement over the industry baseline—while sustaining 19.8% at \$500M. It is the only policy that can support 15%+ carry at any meaningful scale. At \$100M, the ALG reduces CVaR_{95} by 56% and max drawdown by 71% vs. the industry baseline, while achieving the highest Sharpe ratio at every scale. The smoothed ALG-S variant, which accounts for leverage-transition costs, trails the myopic ALG by less than 0.4 pp, confirming the robustness of the mechanism. The ALG requires only current-epoch state and a closed-form computation, making it immediately implementable as an on-chain governor contract.

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